

► PROBLEMS, SECTION 4

For each of the following numbers, first visualize where it is in the complex plane. With a little practice you can quickly find x , y , r , θ in your head for these simple problems. Then

plot the number and label it in five ways as in Figure 3.3. Also plot the complex conjugate of the number.

1. $1 + i$
2. $i - 1$
3. $1 - i\sqrt{3}$
4. $-\sqrt{3} + i$
5. $2i$
6. $-4i$
7. -1
8. 3
9. $2i - 2$
10. $2 - 2i$
11. $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
12. $4\left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3}\right)$
13. $\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$
14. $2\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$
15. $\cos \pi - i \sin \pi$
16. $5(\cos 0 + i \sin 0)$
17. $\sqrt{2}e^{-i\pi/4}$
18. $3e^{i\pi/2}$
19. $5(\cos 20^\circ + i \sin 20^\circ)$
20. $7(\cos 110^\circ - i \sin 110^\circ)$

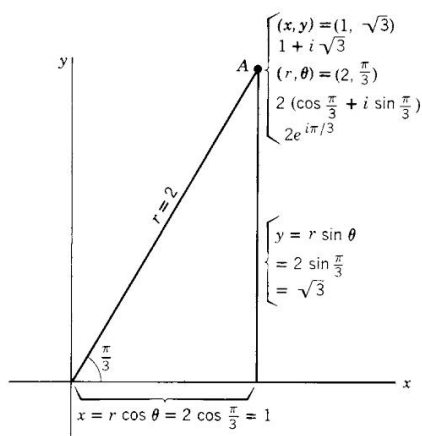


Figure 3.3

【P51】 需要注意写出五种形式，并同时画出给定复数及其共轭复数。

Problem 1

	x	y	r	θ	
4.10	2	-2	$2\sqrt{2}$	$-\pi/4$	$z = x + iy = re^{i\theta}$
4.11	$\sqrt{3}$	1	2	$\pi/6$	$\bar{z} = x - i\theta = re^{-i\theta}$
4.17	1	-1	$\sqrt{2}$	$-\pi/4$	You should plot both
4.18	0	3	3	$\pi/2$	z and $\bar{z}$ in the figure

► PROBLEMS, SECTION 5

First simplify each of the following numbers to the $x + iy$ form or to the $re^{i\theta}$ form. Then plot the number in the complex plane.

1. $\frac{1}{1+i}$

2. $\frac{1}{i-1}$

3. i^4

4. $i^2 + 2i + 1$

5. $(i + \sqrt{3})^2$

6. $\left(\frac{1+i}{1-i}\right)^2$

7. $\frac{3+i}{2+i}$

8. $1.6 - 2.7i$

9. $25e^{2i}$ Careful! The angle is 2 radians.

10. $\frac{3i-7}{i+4}$ Careful! Not $3-7i$

11. $17 - 12i$

12. $3(\cos 28^\circ + i \sin 28^\circ)$

13. $5\left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}\right)$

14. $2.8e^{-i(1.1)}$

15. $\frac{5-2i}{5+2i}$

16. $\frac{1}{0.5(\cos 40^\circ + i \sin 40^\circ)}$

【P52】 需要注意题目要求化成指定的形式，以及需要绘制在复数平面中

Problem 2

	x	y	r	θ
5.5	2	$2\sqrt{3}$	4	$\pi/3$
5.6	-1	0	1	π
5.15	$21/29$	$-20/29$	1	$-43.6^\circ = -0.76$
5.16	1.53	-1.29	2	$-40^\circ = -0.698$

► PROBLEMS, SECTION 5

Solve for all possible values of the real numbers x and y in the following equations.

35. $x + iy = 3i - 4$

36. $2ix + 3 = y - i$

37. $x + iy = 0$

38. $x + iy = 2i - 7$

39. $x + iy = y + ix$

40. $x + iy = 3i - ix$

41. $(2x - 3y - 5) + i(x + 2y + 1) = 0$

42. $(x + 2y + 3) + i(3x - y - 1) = 0$

43. $(x + iy)^2 = 2ix$

44. $x + iy = (1 - i)^2$

45. $(x + iy)^2 = (x - iy)^2$

46. $\frac{x + iy}{x - iy} = -i$

47. $(x + iy)^3 = -1$

48. $\frac{x + iy + 2 + 3i}{2x + 2iy - 3} = i + 2$

49. $|1 - (x + iy)| = x + iy$

50. $|x + iy| = y - ix$

【P54】

Problem 3

$$\begin{aligned}
 5.45 \quad (x+iy)^2 &= (x-iy)^2 \\
 x^2 + i2xy - y^2 &= x^2 - i2xy - y^2 \\
 xy &= \\
 \Rightarrow x=0, y \in \mathbb{R} \quad \text{or} \quad y=0, x \in \mathbb{R}
 \end{aligned}$$

$$\begin{aligned}
 5.50 \quad \underline{|x+iy|} &= y - ix \\
 \text{Re}[z] &\geq 0 \\
 \Rightarrow x=0, y \geq 0
 \end{aligned}$$

► PROBLEMS, SECTION 5

Describe geometrically the set of points in the complex plane satisfying the following equations.

51. $|z| = 2$

52. $\operatorname{Re} z = 0$

53. $|z - 1| = 1$

54. $|z - 1| < 1$

55. $z - \bar{z} = 5i$

56. angle of $z = \frac{\pi}{2}$

57. $\operatorname{Re}(z^2) = 4$

58. $\operatorname{Re} z > 2$

59. $|z + 3i| = 4$

60. $|z - 1 + i| = 2$

61. $\operatorname{Im} z < 0$

62. $|z + 1| + |z - 1| = 8$

63. $z^2 = \bar{z}^2$

64. $z^2 = -\bar{z}^2$

65. Show that $|z_1 - z_2|$ is the distance between the points z_1 and z_2 in the complex plane. Use this result to identify the graphs in Problems 53, 54, 59, and 60 without computation.

【P55】

Problem 4

$$5.59 \quad |z + 3i| = 4$$

$$|x + i(y+3)| = 4$$

$$\Rightarrow x^2 + (y+3)^2 = 16$$

Circle, center at $(0, -3)$, $r = 4$

$$5.62 \quad |z+1| + |z-1| = 8$$

$$|x+1+iy| + |x-1+iy| = 8$$

$$\sqrt{(x+1)^2 + y^2} + \sqrt{(x-1)^2 + y^2} = 8$$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{15} = 1$$

Ellipse, foci at $(1, 0)$ and $(-1, 0)$ semi-major axis is 4

► PROBLEMS, SECTION 11

1. Define $\sin z$ and $\cos z$ by their power series. Write the power series for e^{iz} . By comparing these series obtain the definition (11.4) of $\sin z$ and $\cos z$.
2. Solve the equations $e^{i\theta} = \cos \theta + i \sin \theta$, $e^{-i\theta} = \cos \theta - i \sin \theta$, for $\cos \theta$ and $\sin \theta$ and so obtain equations (11.3).

Find each of the following in rectangular form $x + iy$ and check your results by computer. Remember to save time by doing as much as you can in your head.

$$3. \quad e^{-(i\pi/4) + \ln 3}$$

$$4. \quad e^{3 \ln 2 - i\pi}$$

$$5. \quad e^{(i\pi/4) + (\ln 2)/2}$$

$$6. \quad \cos(i \ln 5)$$

$$7. \quad \tan(i \ln 2)$$

$$8. \quad \cos(\pi - 2i \ln 3)$$

$$9. \quad \sin(\pi - i \ln 3)$$

$$10. \quad \sin(i \ln i)$$

In the following integrals express the sines and cosines in exponential form and then integrate to show that:

$$11. \quad \int_{-\pi}^{\pi} \cos 2x \cos 3x \, dx = 0$$

$$12. \quad \int_{-\pi}^{\pi} \cos^2 3x \, dx = \pi$$

$$13. \quad \int_{-\pi}^{\pi} \sin 2x \sin 3x \, dx = 0$$

$$14. \quad \int_0^{2\pi} \sin^2 4x \, dx = \pi$$

$$15. \quad \int_{-\pi}^{\pi} \sin 2x \cos 3x \, dx = 0$$

$$16. \quad \int_{-\pi}^{\pi} \sin 3x \cos 4x \, dx = 0$$

Evaluate $\int e^{(a+ib)x} dx$ and take real and imaginary parts to show that:

$$17. \quad \int e^{ax} \cos bx \, dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$$

$$18. \quad \int e^{ax} \sin bx \, dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$$

Problem 5

$$11.6 \quad \cos(i \ln 5) = \frac{e^{i \cdot i \ln 5} + e^{-i \cdot i \ln 5}}{2} = \frac{13}{5}$$

$$11.10 \quad \sin(i \ln i) = \frac{e^{i \cdot i \ln i} - e^{-i \cdot i \ln i}}{2i} = -1$$

11.17

$$\begin{aligned} \int e^{ax} \cos bxdx &= \frac{1}{a} \int \cos bxd e^{ax} = \frac{1}{a} \cos bxe^{ax} + \frac{b}{a} \int e^{ax} \sin bxdx \\ &= \frac{e^{ax}}{a} \cos bx + \frac{b}{a^2} \sin bxe^{ax} - \frac{b^2}{a^2} \int e^{ax} \cdot \cos bxdx \\ &\Rightarrow \int e^{ax} \cos bxdx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} \end{aligned}$$

Find each of the following in the $x + iy$ form and check your answers by computer.

29. $\cosh 2\pi i$

30. $\tanh \frac{3\pi i}{4}$

31. $\sinh \left(\ln 2 + \frac{i\pi}{3} \right)$

32. $\cosh \left(\frac{i\pi}{2} - \ln 3 \right)$

33. $\tan i$

34. $\sin \frac{i\pi}{2}$

【P71】

Problem 6

$$12.33 \quad \tan i = \frac{\sin i}{\cos i} = \frac{\frac{e^{ii} - e^{-ii}}{2i}}{\frac{e^{ii} + e^{-ii}}{2}} = \frac{e - e^{-1}}{e + e^{-1}} i = 0.762i$$

$$12.34 \quad \sin \frac{i\pi}{2} = \frac{e^{i \cdot \frac{i\pi}{2}} - e^{-i \cdot \frac{i\pi}{2}}}{2i} = \frac{e^{-\frac{\pi}{2}} - e^{\frac{\pi}{2}}}{2} i = -2.301i$$

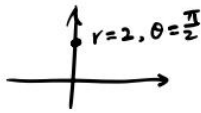
► PROBLEMS, SECTION 14

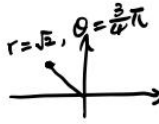
Evaluate each of the following in $x + iy$ form, and compare with a computer solution.

- | | | |
|--|---------------------------------|---|
| 1. $\ln(-e)$ | 2. $\ln(-i)$ | 3. $\ln(i + \sqrt{3})$ |
| 4. $\ln(i - 1)$ | 5. $\ln(-\sqrt{2} - i\sqrt{2})$ | 6. $\ln\left(\frac{1-i}{\sqrt{2}}\right)$ |
| 7. $\ln\left(\frac{1+i}{1-i}\right)$ | 8. $i^{2/3}$ | 9. $(-1)^i$ |
| 10. $i^{\ln i}$ | 11. 2^i | 12. i^{3+i} |
| 13. $i^{2i/\pi}$ | 14. $(2i)^{1+i}$ | 15. $(-1)^{\sin i}$ |
| 16. $\left(\frac{1+i\sqrt{3}}{2}\right)^i$ | 17. $(i-1)^{i+1}$ | 18. $\cos(2i \ln i)$ |

【P74】

Problem 7

$$\begin{aligned}
 14.14 \quad (2i)^{1+i} &= e^{(1+i)\ln 2i} \\
 &= e^{(1+i)(\ln 2 + i\frac{\pi}{2})} \\
 &= 2e^{-\frac{\pi}{2}} \cdot e^{i(\frac{\pi}{2} + \ln 2)} \\
 &= 2e^{-\frac{\pi}{2}} [\cos(\ln 2 + \frac{\pi}{2}) + i\sin(\ln 2 + \frac{\pi}{2})] \\
 &= 2e^{-\frac{\pi}{2}} [-\sin(\ln 2) + i\cos(\ln 2)] \\
 &= -0.2657 + 0.3198i
 \end{aligned}$$


$$\begin{aligned}
 14.17 \quad (i-1)^{i+1} &= e^{(i+1)\ln(i-1)} \\
 &= e^{(i+1)(\ln\sqrt{2} + i\frac{3\pi}{4})} \\
 &= \sqrt{2} e^{-\frac{3\pi}{4}} e^{i(\ln\sqrt{2} + \frac{3\pi}{4})} \\
 &= -0.121 + 0.057i
 \end{aligned}$$


► PROBLEMS, SECTION 16

11. Prove that

$$\begin{aligned}
 \cos \theta + \cos 3\theta + \cos 5\theta + \cdots + \cos(2n-1)\theta &= \frac{\sin 2n\theta}{2\sin \theta}, \\
 \sin \theta + \sin 3\theta + \sin 5\theta + \cdots + \sin(2n-1)\theta &= \frac{\sin^2 n\theta}{\sin \theta}.
 \end{aligned}$$

Hint: Use Euler's formula and the geometric progression formula.

【P79】

Problem 8

$$16.11 \quad \sum_{k=1}^n e^{i(2k-1)\theta} = \sum_{k=1}^n [\cos(2k-1)\theta + i\sin(2k-1)\theta]$$

$$= \sum_{k=1}^n \cos(2k-1)\theta + i \sum_{k=1}^n \sin(2k-1)\theta$$

$$\cos\theta + \cos 3\theta + \cos 5\theta + \dots + \cos(2n-1)\theta = \operatorname{Re} \left[\sum_{k=1}^n e^{i(2k-1)\theta} \right]$$

$$\sin\theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta = \operatorname{Im} \left[\sum_{k=1}^n e^{i(2k-1)\theta} \right]$$

$$\begin{aligned} \sum_{k=1}^n e^{i(2k-1)\theta} &= \frac{e^{i\theta}(1 - e^{i2n\theta})}{1 - e^{i2\theta}} = \frac{1 - e^{i2n\theta}}{e^{-i\theta} - e^{i\theta}} \\ &= \frac{-2i[1 - (\cos 2n\theta + i\sin 2n\theta)]}{(e^{i\theta} - e^{-i\theta})2i} \\ &= \frac{\cos 2n\theta + i\sin 2n\theta - 1}{2i\sin\theta} \\ &= \frac{\sin 2n\theta}{2\sin\theta} + \frac{\cos 2n\theta - 1}{2i\sin\theta} \\ &= \frac{\sin 2n\theta}{2\sin\theta} + \frac{1 - 2\sin^2 n\theta - 1}{2i\sin\theta} \end{aligned}$$

$$= \frac{\sin 2n\theta}{2\sin\theta} + i \frac{\sin^2 n\theta}{\sin\theta}$$

$$\Rightarrow \sum_{k=1}^n \cos(2k-1)\theta = \operatorname{Re} \left[\sum_{k=1}^n e^{i(2k-1)\theta} \right] = \frac{\sin 2n\theta}{2\sin\theta}$$

$$\sum_{k=1}^n \sin(2k-1)\theta = \operatorname{Im} \left[\sum_{k=1}^n e^{i(2k-1)\theta} \right] = \frac{\sin^2 n\theta}{\sin\theta}$$

Solving with mathematical induction is OK but not recommended

25. (a) Show that $\overline{\cos z} = \cos \bar{z}$.

(b) Is $\overline{\sin z} = \sin \bar{z}$?

【P81】b 问有一些同学做错了，没有注意分母中的 i。

Problem 9

$$\begin{aligned}\text{(a)} \quad \overline{\cos z} &= \frac{\overline{e^{iz} + e^{-iz}}}{2} \\ &= \frac{\overline{e^{iz}} + \overline{e^{-iz}}}{2} \\ &= \frac{e^{-i\bar{z}} + e^{i\bar{z}}}{2} \\ &= \cos \bar{z}\end{aligned}$$

f, g are complex numbers
 $z = f + ig \Rightarrow \bar{z} = \bar{f} - i\bar{g}$
 $z = \frac{f}{g} \Rightarrow \bar{z} = \frac{\bar{f}}{\bar{g}}$
 $\overline{e^{iz}} = e^{-i\bar{z}}$

$$\text{(b)} \quad \overline{\sin z} = \frac{\overline{e^{iz} - e^{-iz}}}{2i} = \frac{e^{-i\bar{z}} - e^{i\bar{z}}}{-2i} = \sin \bar{z}$$